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Title of Lecture: Physics of Ultrasound

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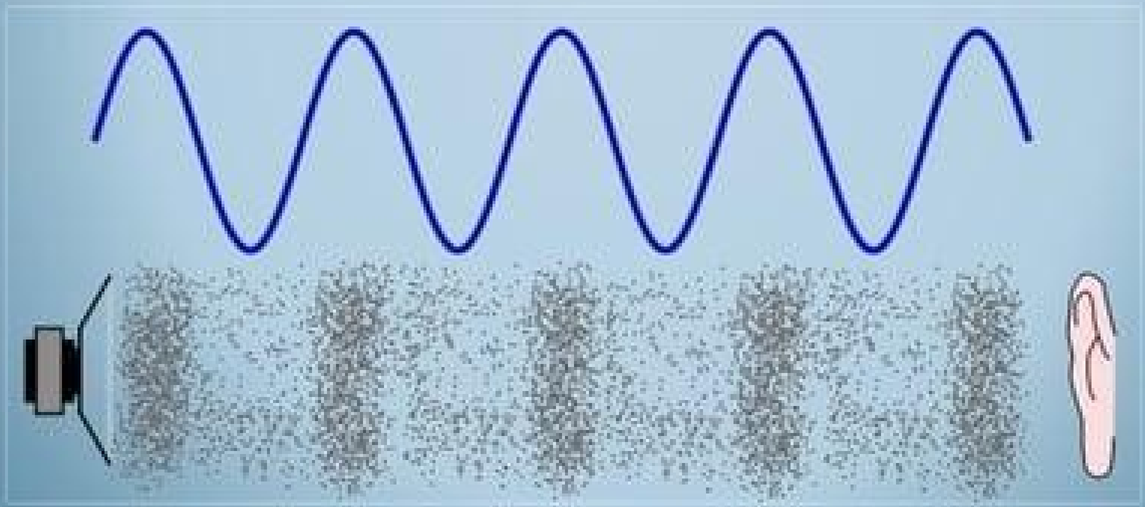
What is Sound ?

- Mechanical and Longitudinal waves wave that can transfer a distance using a media.
- Cannot travel through Vacuum.

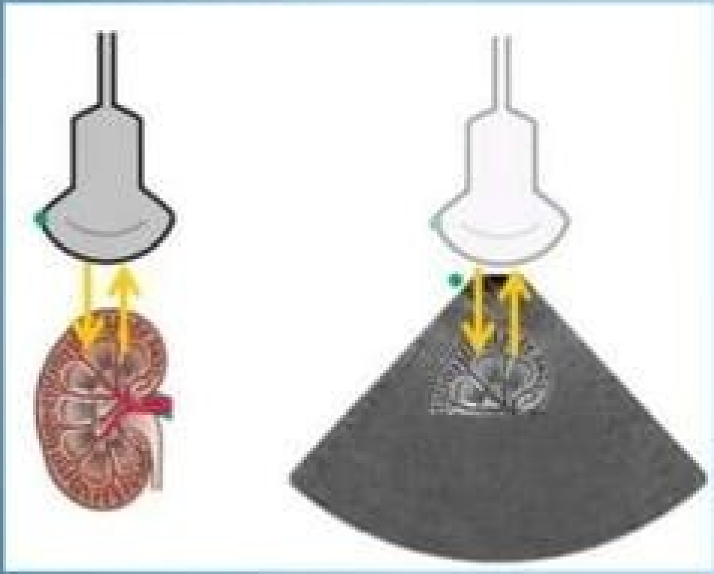


What is Ultrasound?

- Ultrasound is a mechanical, longitudinal wave with a frequency exceeding the upper limit of human hearing, which is 20,000 Hz or 20 kHz.
- Typically at 2 – 20 Mhz.
- Average speed of ultrasound in body is 1540m/sec

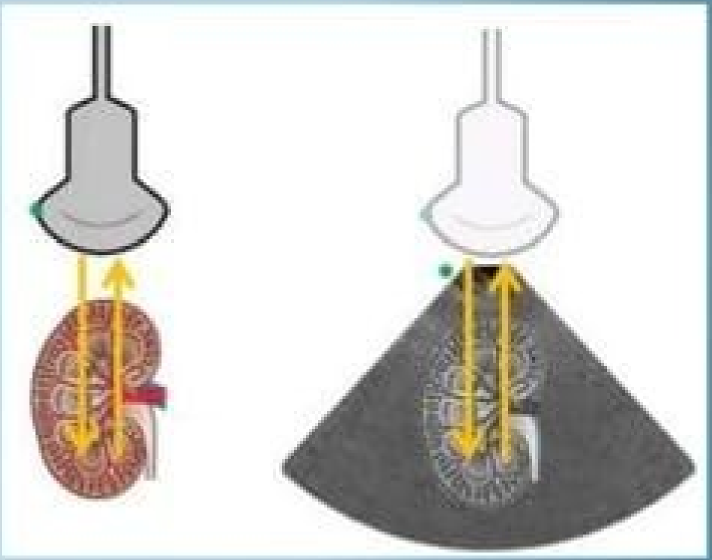
Velocity

Near Field Imaging



Tissues closer appear on top and faster the waves return

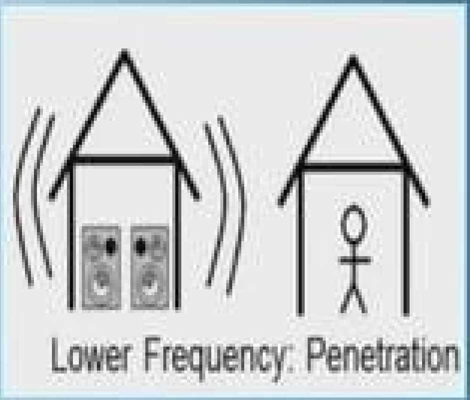
Far Field Imaging



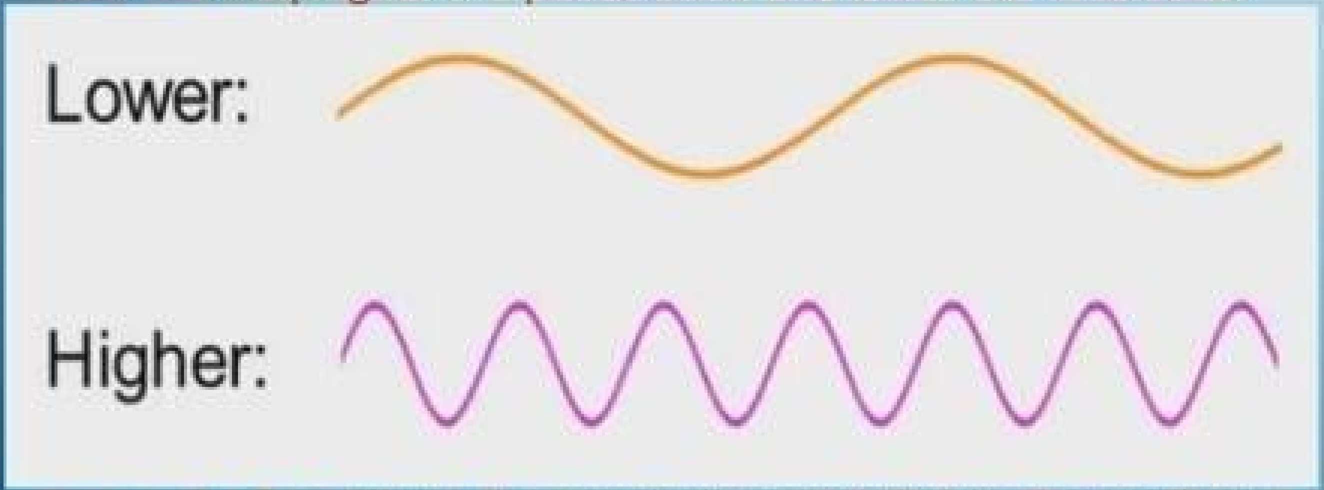
Tissues further appear at the bottom & slower the waves return

Frequency

- Number of cycles per second
- Units are Hertz
- Ultrasound imaging frequency range
  2-20Mhz



Low the freq higher the penetration and lower the resolution

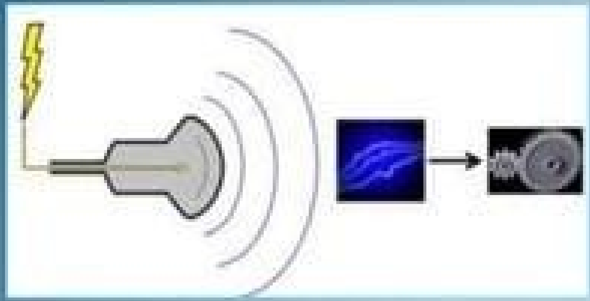


Higher the freq Lower the penetration and Higher the resolution

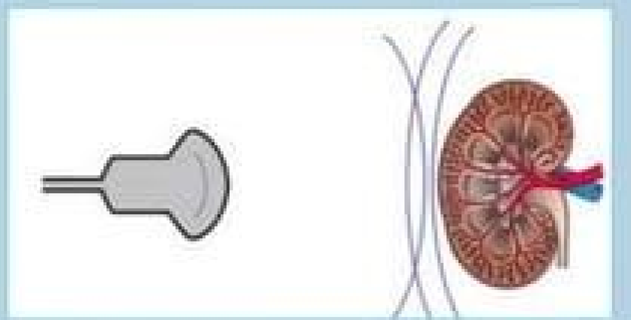
How Ultrasound Works

????

Piezoelectric Effect of Ultrasound



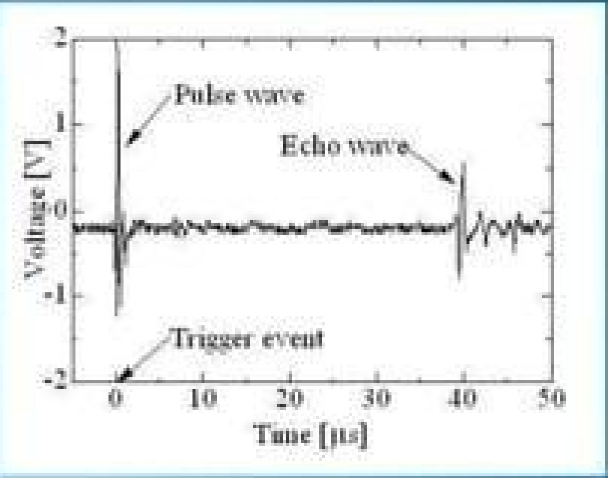
1. Electrical Energy converted to Sound waves



2. The Sound waves are reflected by tissues

3. Reflected Sound waves are converted to electrical signals and later to Image

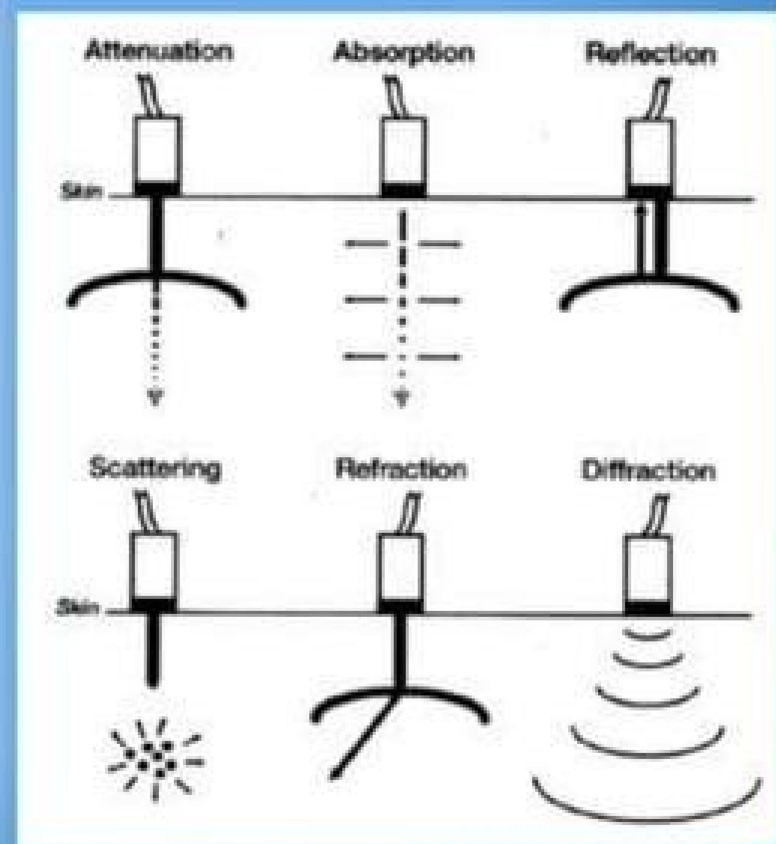
Pulse-Echo Method



- Ultrasound transducer produces "pulses" of ultrasound waves
- These waves travel within the body and interact with various tissues
- The reflected waves return to the transducer and are processed by the ultrasound machine
- An image which represents these reflections is formed on the monitor

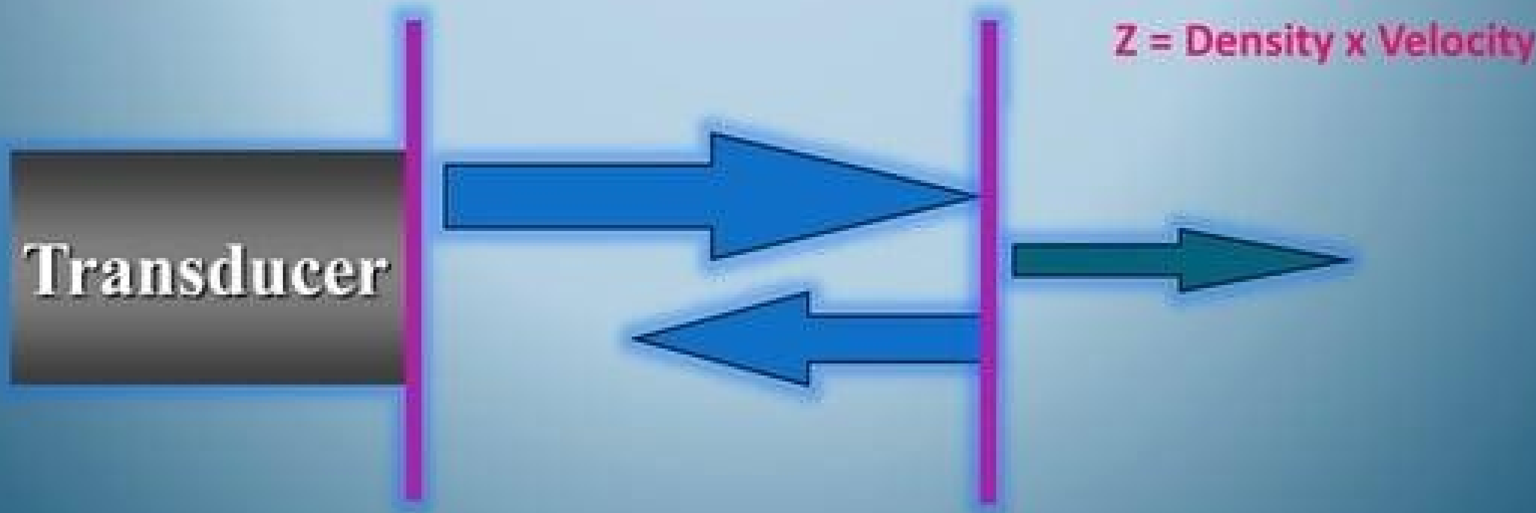
Interactions of Ultrasound with tissue

- Reflection
- Transmission
- Attenuation
- Scattering



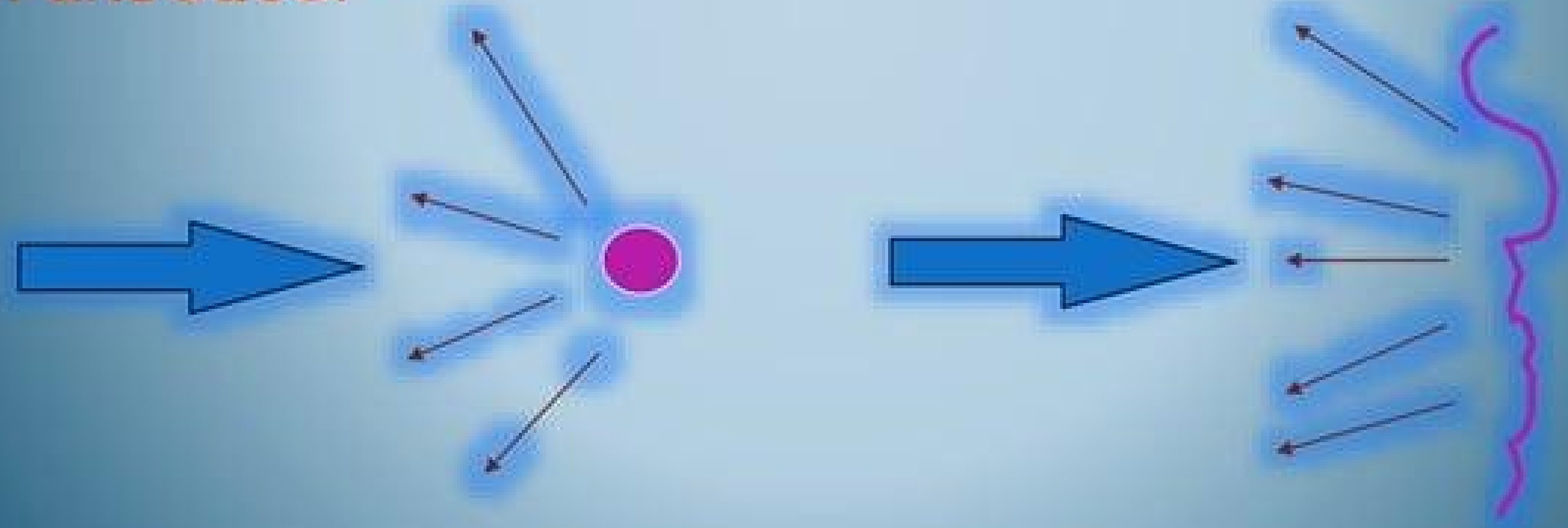
Reflection

- Occurs at a boundary between 2 adjacent tissues or media
- The amount of reflection depends on differences in acoustic impedance (z) between media
- The ultrasound image is formed from reflected echoes



Scattering

- Redirection of sound in several directions
- Caused by interaction with small reflector or rough surface
- Only portion of sound wave returns to transducer

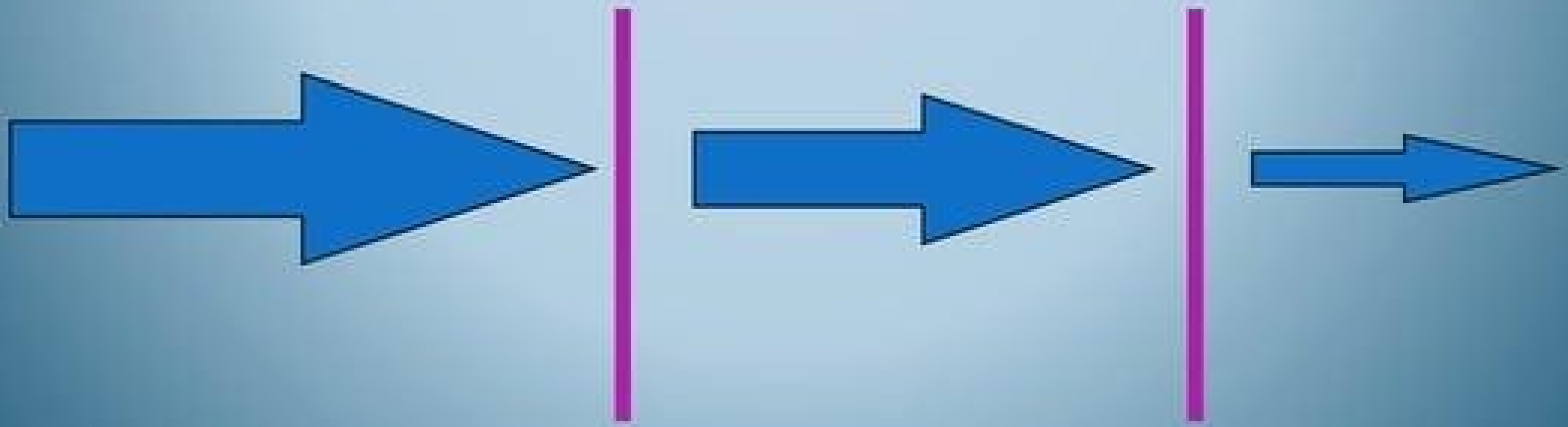


Transmission

- Not all the sound wave is reflected, some continues deeper into the body
- These waves will reflect from deeper tissue structures

Attenuation

- The deeper the wave travels in the body, the weaker it becomes
- The amplitude of the wave decreases with increasing depth



Echogenicity (caused by Reflection)

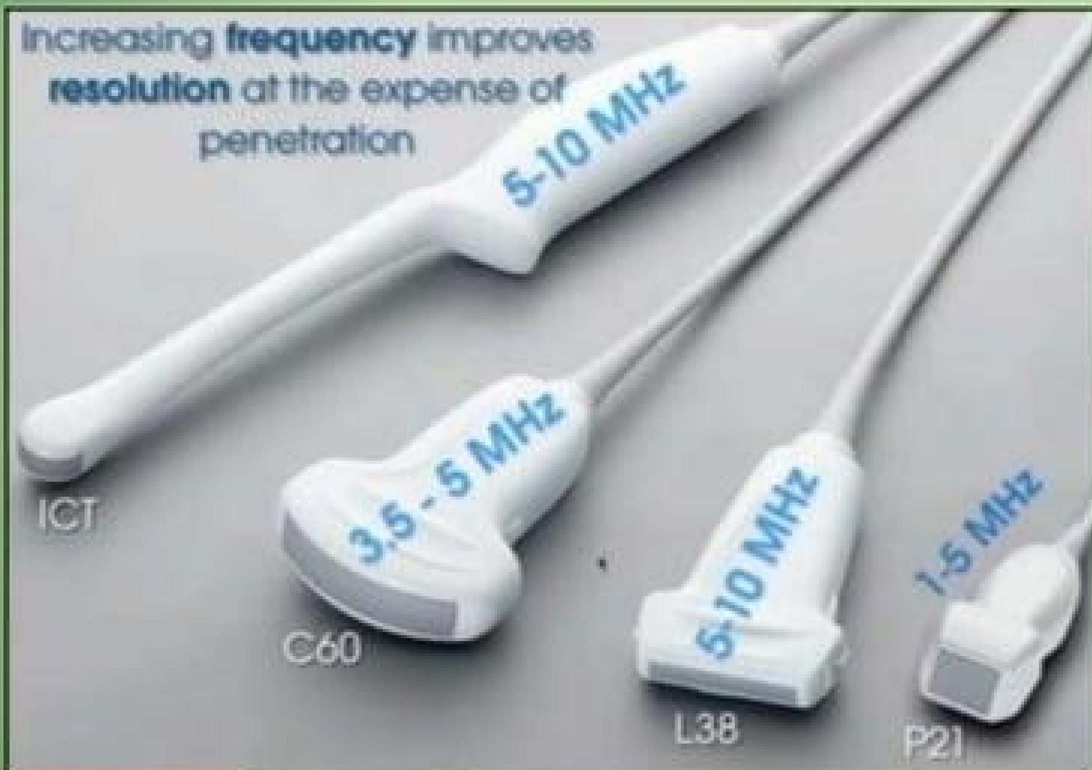
Ane-Echoic

Hypeo-Echoic

Hyper-Echoic

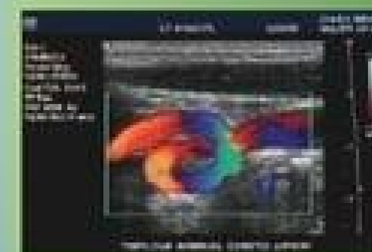
Probes

Range is ideal to switch between General, High and low Resolution



Curvilinear or Abdom

Linear or Vascular



Phased Array or Cardiac

Thank you